

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	XXXXXX
Project Name	Human Development Index (HDI) Predictor
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	HDI Prediction	Predicts Human Development Index using the trained Gradient Boosting model	ML Model	Done	High
2	Data Preprocessing	Cleans dataset, removes missing values, converts data types, and prepares features	Data Processing	Done	High
3	Interactive Dashboard	Displays prediction results, gauge chart, AI insights, and statistics	React Frontend	Done	High
4	Input Validation	Validates all user inputs and handles invalid requests gracefully	Flask Backend	Done	High
5	REST API Integration	Connects React frontend with Flask backend using Axios	API Layer	Done	High
6	Prediction History	Stores and displays recent prediction records using browser local storage	Frontend	Done	High
7	CSV & PDF Export	Allows users to export prediction history and results	Export Module	Done	Medium
8	AI Insights	Generates category-based insights and recommendations from prediction results	Analytics Module	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (Prediction, Preprocessing, Dashboard, Validation, REST API)

Additional / Nice-to-Have Features	3 (History, CSV/PDF Export, AI Insights)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Dashboard, Prediction Form, Gauge Chart, AI Insights
2	Backend / Logic	3	Flask API, ML Prediction Engine, Input Validation
3	Database / Storage	1	Local Storage for Prediction History
4	API / Integration	1	Axios REST API between React and Flask
5	Security / Authentication	1	Input Validation and Error Handling (<i>User authentication planned for future enhancement</i>)